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GEOSPATIAL PORTFOLIO

CASE STUDY / SPATIAL DECISION SUPPORT

Parcel Locker Site Suitability Analysis

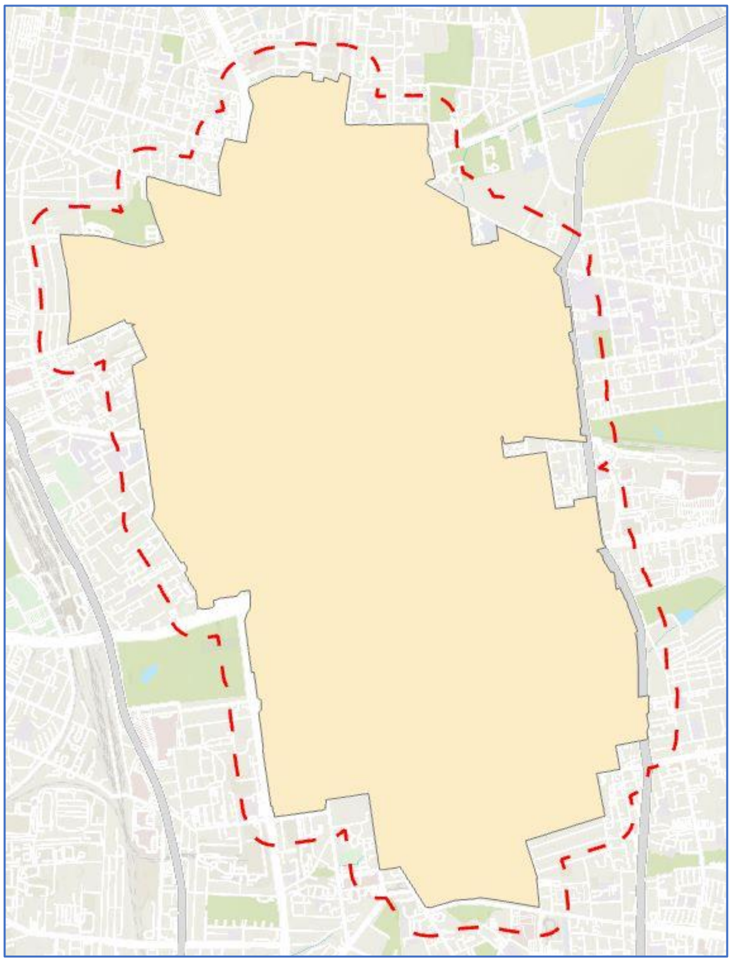
Multi-Criteria Location Analysis for Łódź, Poland

Identifying candidate parcel-locker locations by combining accessibility, demand, competition and spatial constraints.

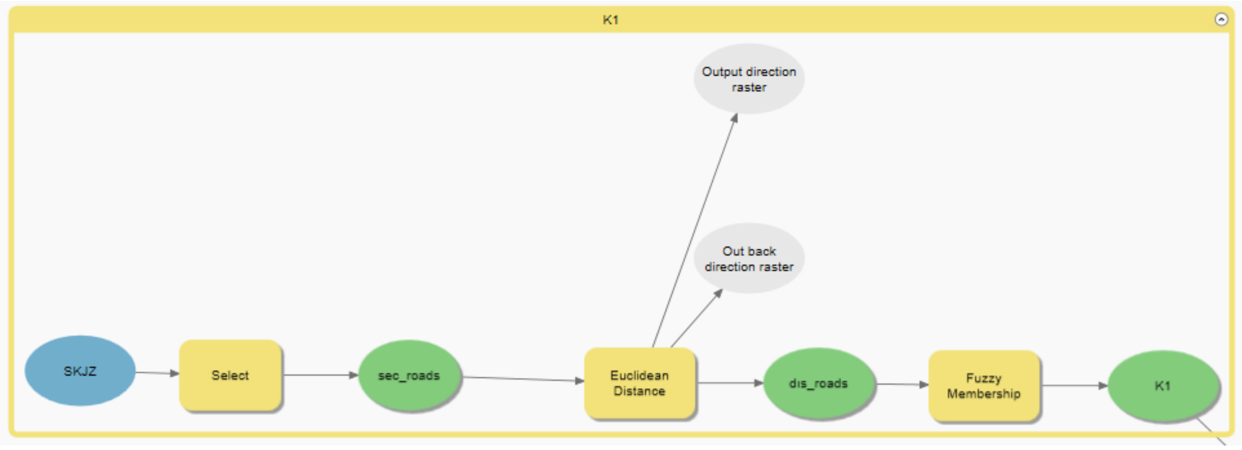
ArcGIS Pro · ModelBuilder · AHP · Weighted Sum · Network Analysis

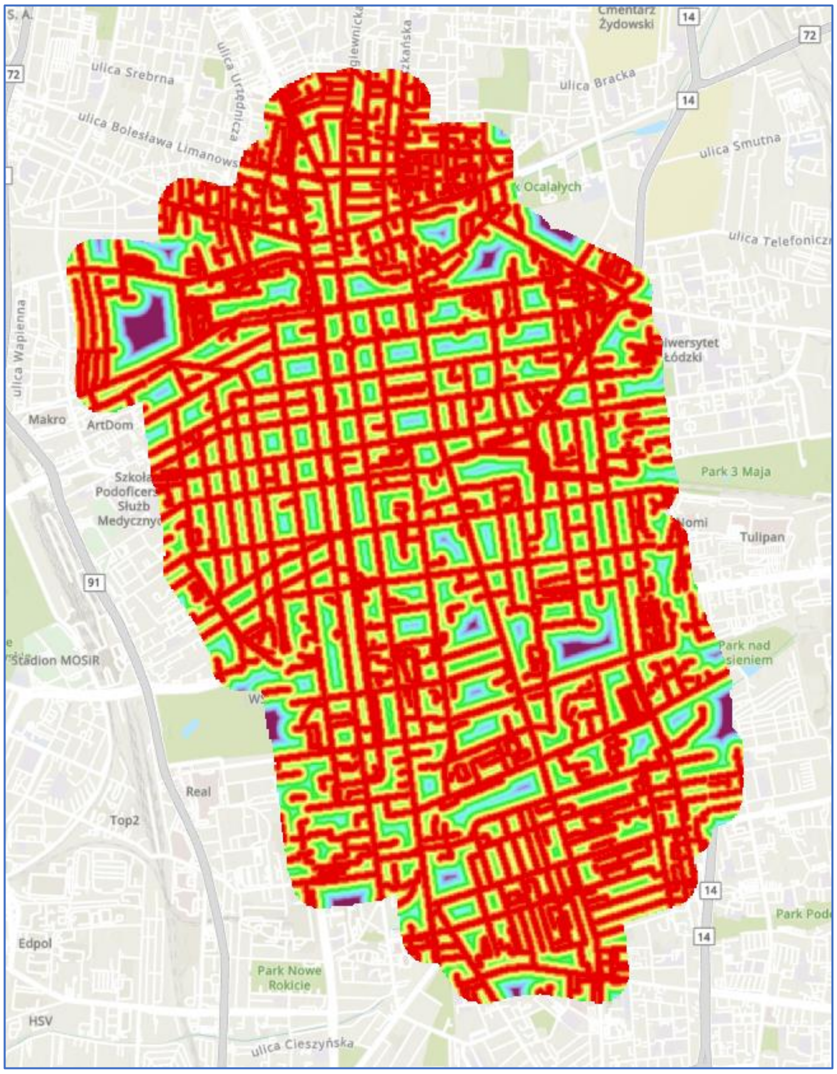
The efficient distribution of parcels entails identifying suitable locations for parcel machines. The location design process involves comprehensive evaluations to locate optimal spots for automating lockers or terminals that fit an area's needs. A priority aim is making last-mile delivery more effective, ensuring ease of access for package recipients while decreasing courier services' workload. Analyzing the distribution of residential regions, business districts, and transit hubs is crucial in a city context. A strategic approach is necessary, according to a study by Piotr Krawiec and Tomasz Kisilowski titled "Location Analysis of Automated Parcel Lockers in a City: A Case Study of Warsaw, Poland" (2020), to make sure that the devices are positioned close to housing developments, shopping malls, and transportation nodes like train stations or bus terminals. Within a predetermined working area, we want to locate the best location for a new inpost parcel locker in the city of Lodz. There are typically a few common standards that must be adhered to or taken into consideration in these initiatives in order accomplish this. In addition to these common criteria, we created our own criteria and made our project more unique. Now let's look at what these criteria are.

We used the Model Builder of Arcgis Pro while doing all the multi-criteria. We used the data we already had. This is our project area and we make every analysis in these certain area.



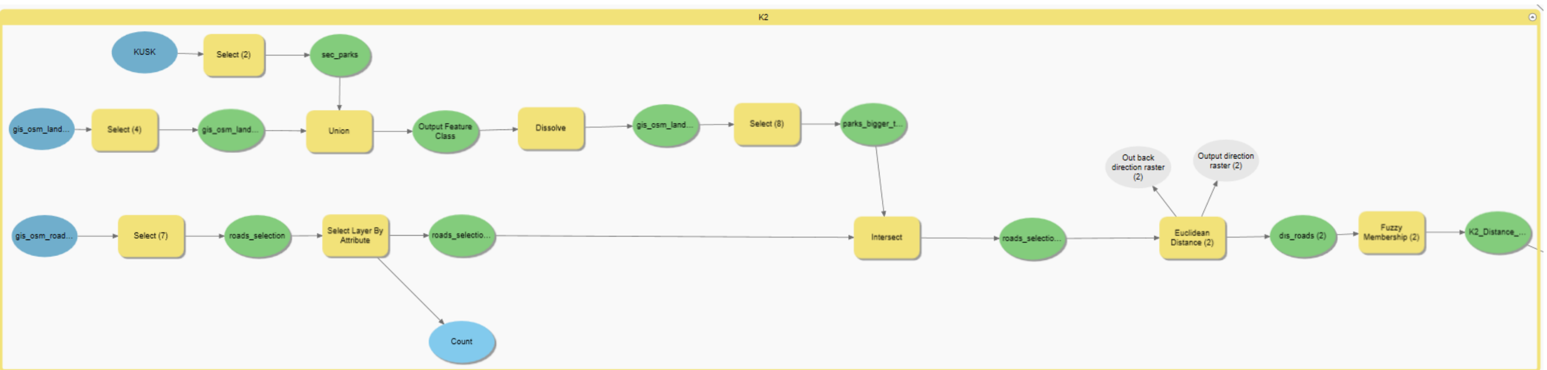
Criteria 1 (Distance to roads): When deciding where to place parcel lockers, one important factor to take into account is the distance to nearby roadways. The ease of access for couriers and recipients is ensured by proximity to main roads, which optimizes delivery routes and cuts down on delivery time. It also offers the essential infrastructure support and aids in reducing traffic congestion problems. Additionally, placing parcel lockers delivery devices close to busy roads improves visibility and security while providing receivers with easy access. The convenience and general efficiency of last-mile delivery can be greatly increased by include this criterion.

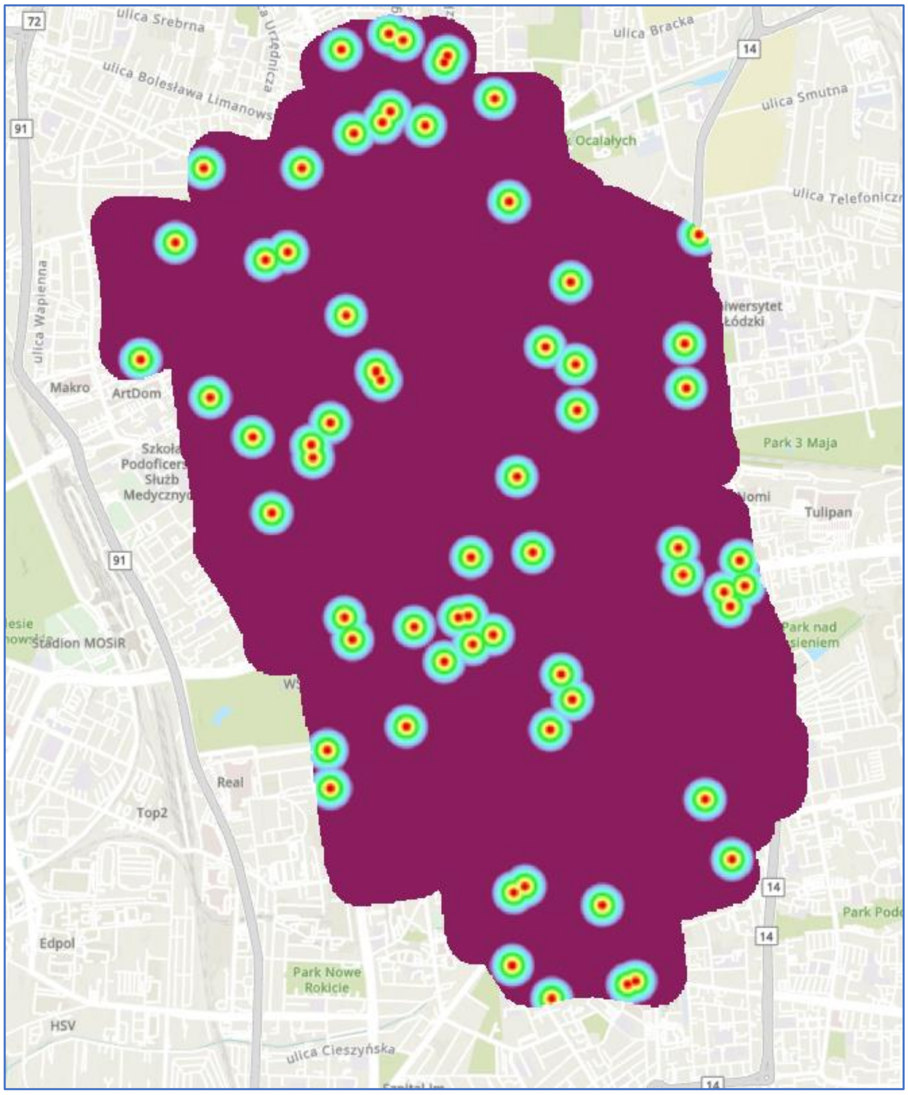




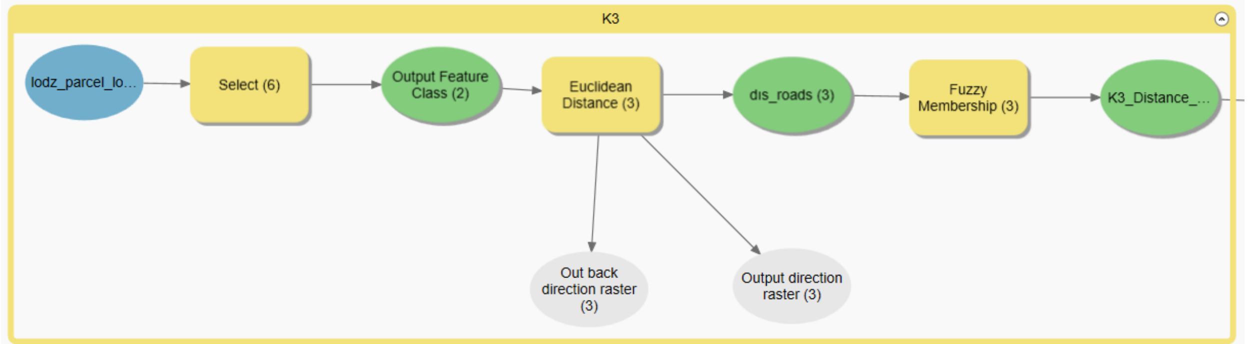
Criteria 2 (Distance to park entrances): When deciding where to put parcel lockers, it's crucial to take the distance to the nearby parking exits into account. For couriers and buyers, accessibility and convenience are provided by proximity to park gates. Due to their close proximity, visitors to the parks may conveniently pick up their items. Additionally, parcel lockers placed close to park entrances take use of the high foot activity in the neighborhood to eliminate the need for customers to make multiple trips to pick up their parcels. Taking into account this criterion has benefits for accessibility, usability, community involvement, and environmental sustainability.

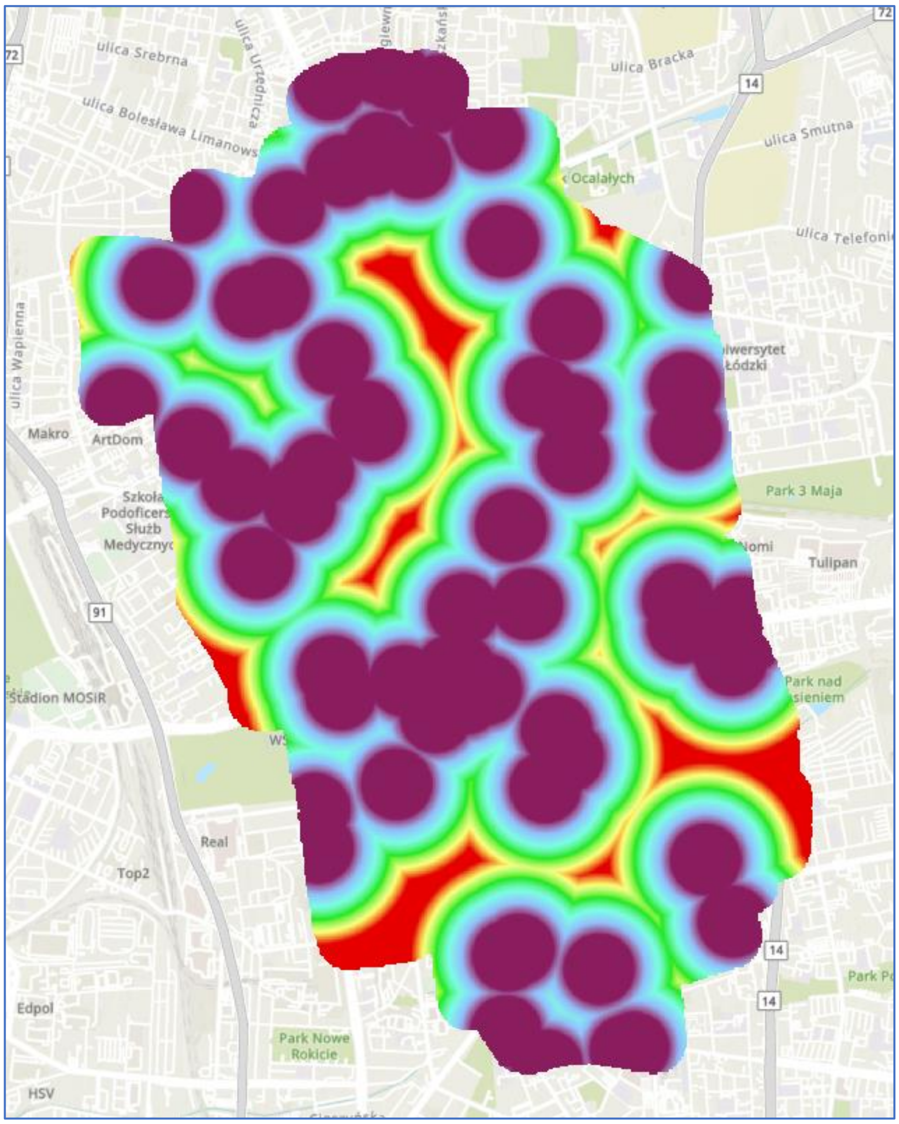
In this criterion, we only base on parks larger than $400m^2$.





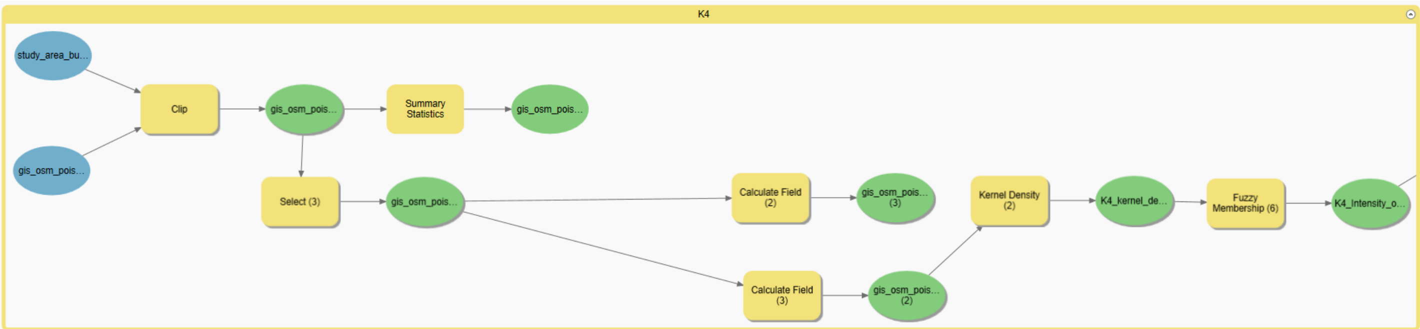
Criteria 3 (Distance to existing parcel lockers): The distance criterion to existing parcel locker must be taken into account while choosing the sites of the parcel lockers in order to provide the greatest distance possible. This criterion's consideration expands the reach and accessibility. In addition, it helps to optimize the best use of each parcel locker and boosts delivery effectiveness. Maximizing the distance to already-existing parcel machines improves last-mile delivery by enhancing coverage, handling, and delivery efficiency.



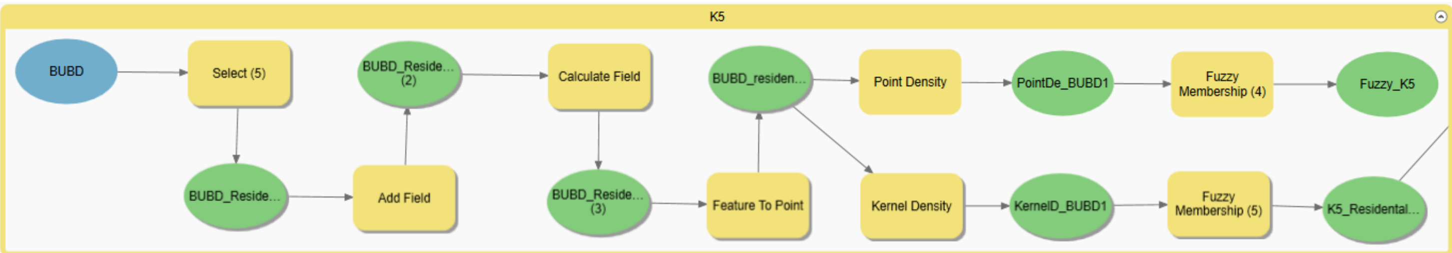


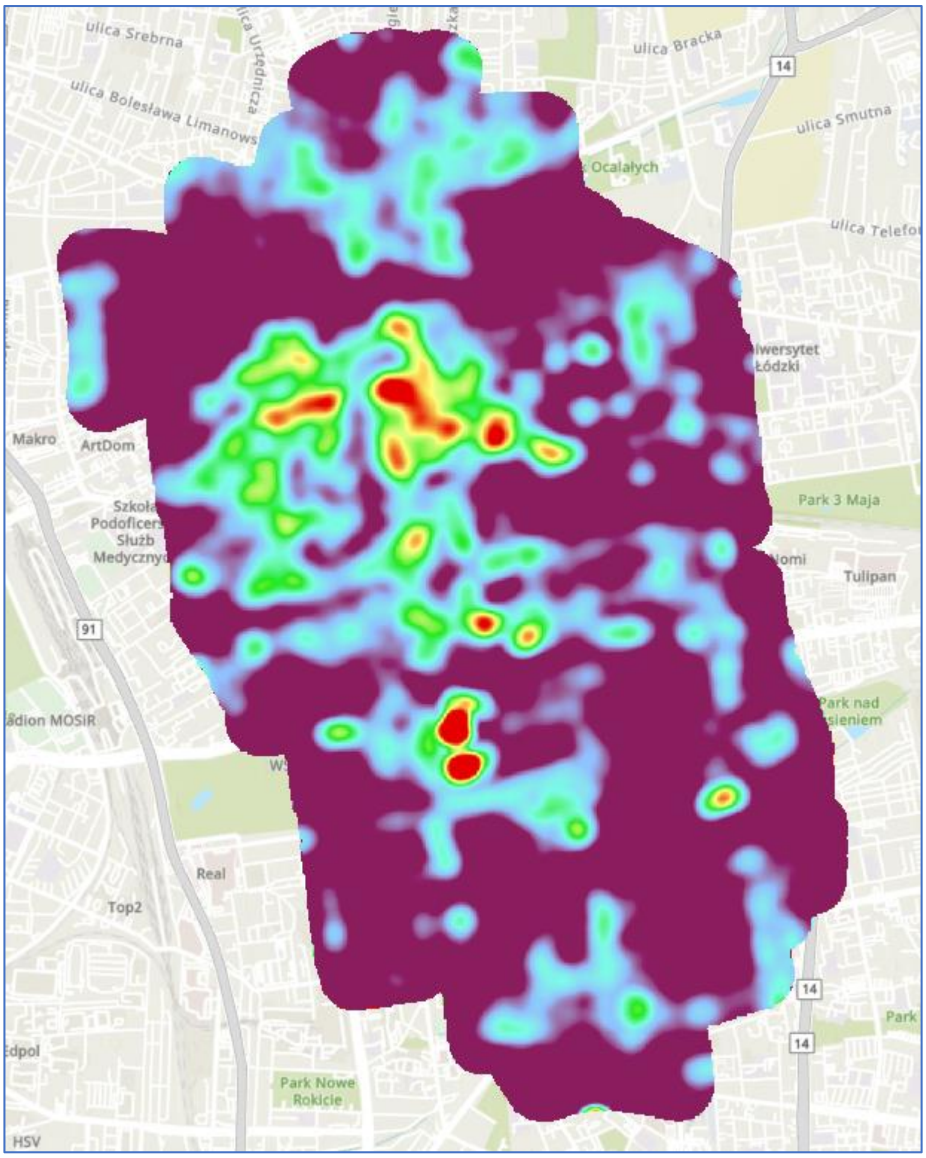
Criteria 4 (Intensity of POI): The density of nearby Points of Interest (POI) should be as high as feasible when choosing locations for parcel lockers. Shopping malls, business hubs and transit hubs are examples of high-intensity POI locations that provide a number of benefits. They allow recipients easy access, increase foot traffic, increase visibility, and present prospective commercial collaborations and marketing opportunities. Locations for parcel lockers can be carefully chosen to maximize convenience, use and overall efficiency in the last-mile delivery process by taking the intensity of POI into account as a factor.

At this stage, we have chosen 10 points that we have determined ourselves, these are for me; became attraction,bakery,clothes, convenience, mall, post box,school, supermarket, town hall, university and vending parking.



Criteria 5(Intensity of residential development): The surrounding residential density requirement is taken into consideration while choosing the locations for the parcel lockers, and efforts are made to choose those areas with the highest density possible. The high housing density makes it simple for purchasers to visit and increases the number of people who could use parcel lockers. Additionally, it promotes community involvement, improves time and cost efficiency and complies with regional requirements for the delivery of parts. The last mile delivery process can be optimized for ease, usability, and operational efficiency by taking residential density into account as a criterion.

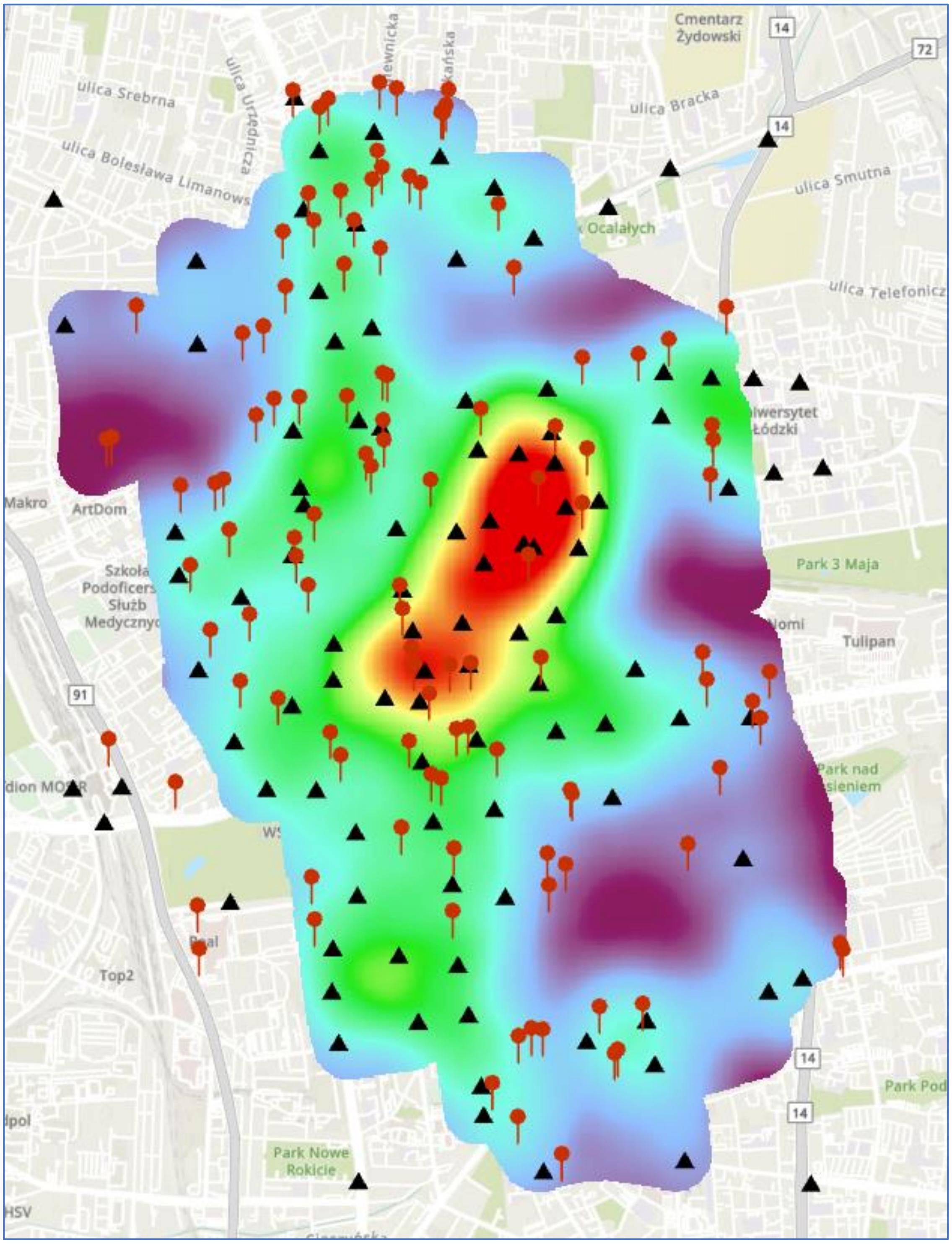




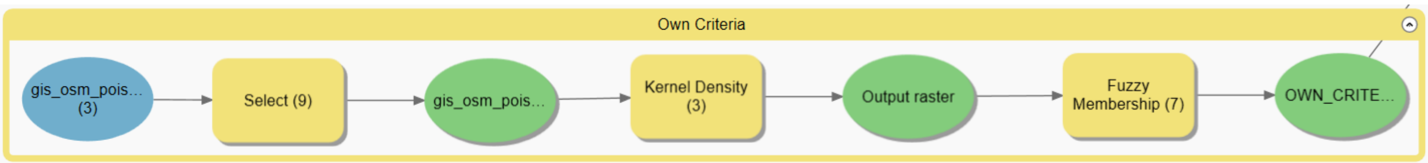
Criteria 6(Own Criteria): I preferred to examine the density of bike rental locations for our own criteria section. I saw these points in the data we had and discovered this after carefully examining it. Between or close to both bike rental locations, there is a parcel locker. According to this criterion, I believed it would be sensible to situate it somewhere without a parcel locker between or close to the bicycle rental locations since renting and riding bicycles is highly widespread in Lodz and it offers a simpler and faster way to transportation.

Triangles = Bicycle Rental Points

Red Pins = Parcel Lockers



You can observe the basis for my selection criteria by looking at the image up above.



After deciding on all of our criteria and taking the necessary steps, it is a process that enables us to arrive at a logical conclusion by comparing the six criteria we have decided, known as criteria weights, in accordance with their importance levels. To determine whether the weight we choose makes sense while we are doing this, we can utilize certain mathematical formula. In ArcGIS Pro that operation calls by ‘’ Weighted Sum ‘’. ArcGIS Pro uses the weighted sum geographic analysis technique. This technique makes it possible to evaluate a point or a region using many criteria. Each criterion is given a weight, which establishes the criteria's order of importance. Each criterion's value is multiplied by its weight before being added together to provide a weighted total for evaluation. This approach is utilized in multi-criteria decision-making procedures and aids in identifying the best possibilities after analysis to choose from. Before this operation we need to make a logical importance level with our criteria. For that we use an excel file which includes our formula (Pairwise Comparison Matrix).

Relative importance	Definition	Explanation
1	Equal importance	Two activities contribute equally to objective
3	Weak importance	Experience and judgment slightly favour one activity over another
5	Strong importance	Experience and judgment strongly favour one activity over anothe
7	Demonstrated importance	One activity is strongly favoured and demonstrated in practice
9	Extreme importance	The evidence favouring one activity over another is of highest possible order of affirmation
2, 4, 6, 8	Intermediate values	When compromise is needed between two adjacent judgments

Criteria	
K1	distance to roads
K2	distance to park entrances
K3	distance to existing parcel lockers
K4	intensity of POIs
K5	intensity of residential development
K6	YOUR OWN CRITERION

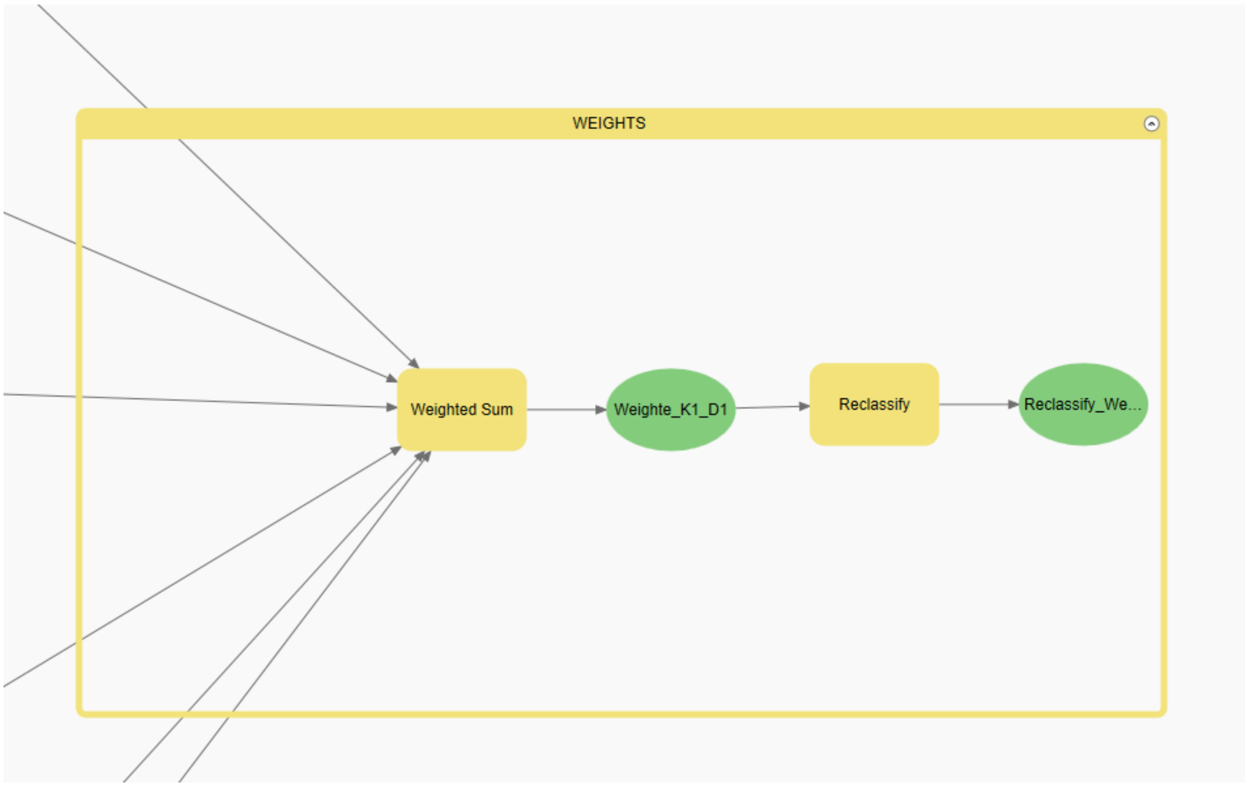
$$CI = \frac{\lambda_{max} - n}{n - 1} \qquad CR = \frac{CI}{RI}$$

	K1	K2	K3	K4	K5	K6
K1	1	5	1	1/3	5	5
K2	1/5	1	1/3	1/5	3	1
K3	1	3	1	1/2	3	3
K4	3	5	2	1	5	5
K5	1/5	1/3	1/3	1/5	1	1
K6	1/5	1	1/3	1/5	1	#
	5 3/5	15 1/3	5	2 13/30	18	##

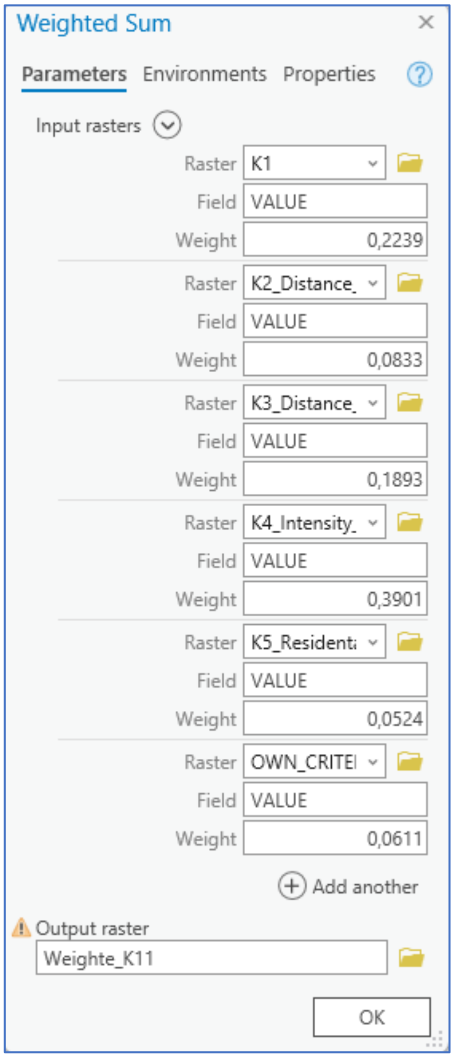
	K1	K2	K3	K4	K5	K6
K1	0,1786	0,3261	0,2000	0,1370	0,2778	0,3125
K2	0,0357	0,0652	0,0667	0,0822	0,1667	0,0625
K3	0,1786	0,1957	0,2000	0,2055	0,1667	0,1875
K4	0,5357	0,3261	0,4000	0,4110	0,2778	0,3125
K5	0,0357	0,0217	0,0667	0,0822	0,0556	0,0625
K6	0,0357	0,0652	0,0667	0,0822	0,0556	0,0625
	1,0000	1,0000	1,0000	1,0000	1,0000	1,0000

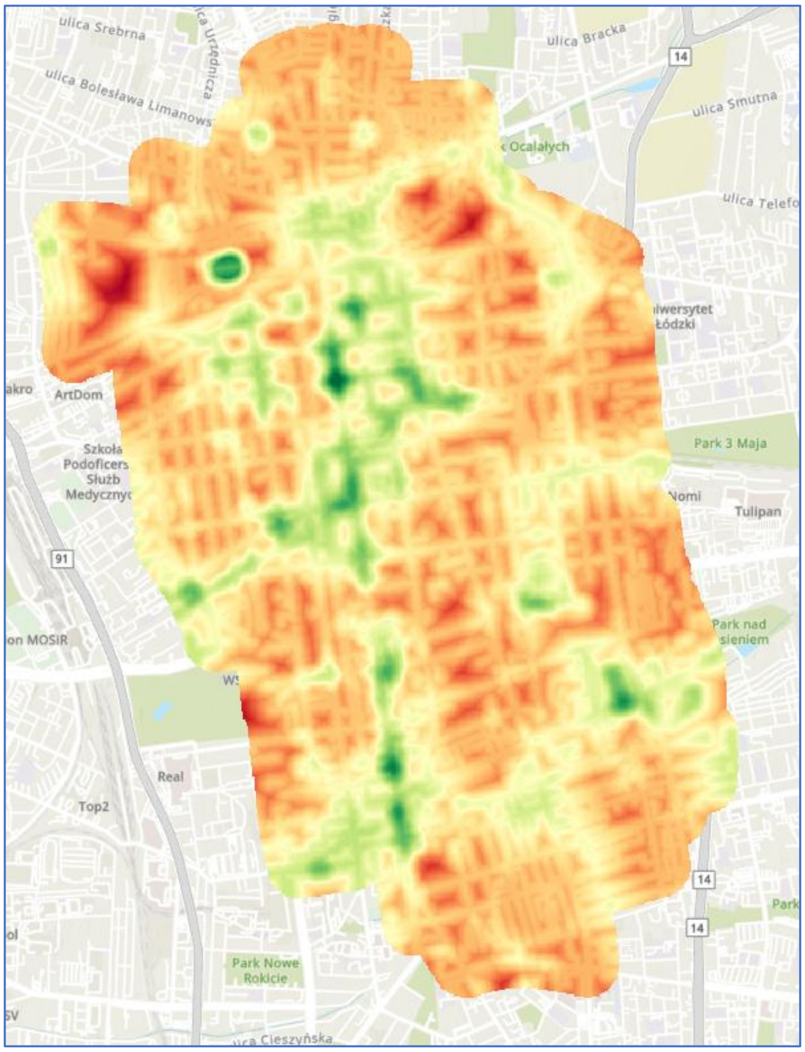
Wj	S	Lambda	
0,2239	1,5269	6,8199	
0,0833	0,4874	5,8514	
0,1893	1,1984	6,3316	
0,3901	2,4240	6,2136	
0,0524	0,3271	6,2455	
0,0611	0,3826	6,2654	
1,0000		6,2879	LambdaMax
	CI=	0,058	
	CR=	0,046	0.1

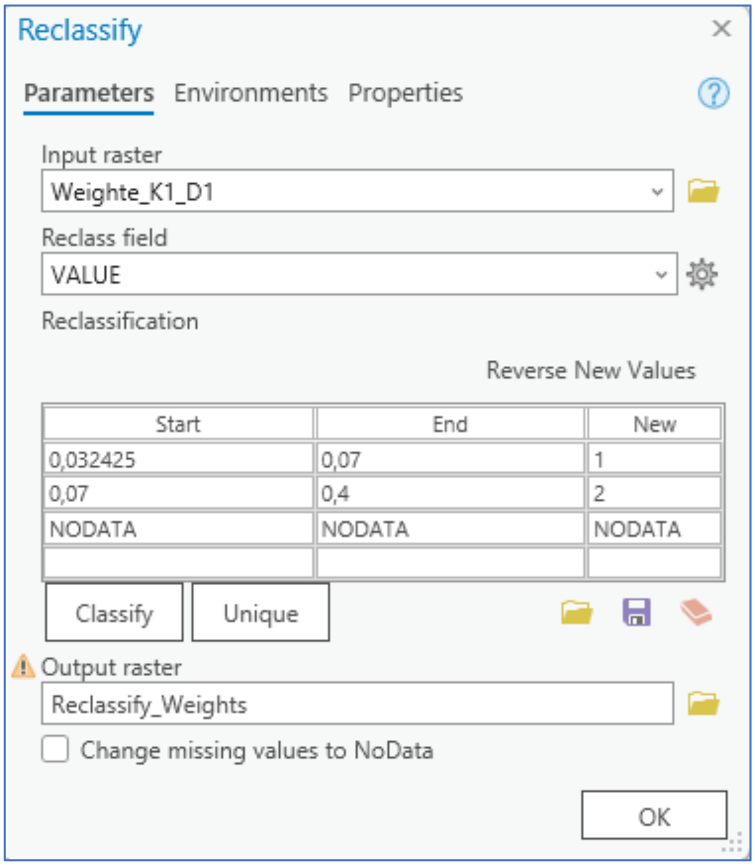
The logicalness of the outcome increases when the CR value decreases below 0.1. The values and outcomes I presented can be seen in the images. With these values we can operate the ‘’ Weighted Sum’’ operation.

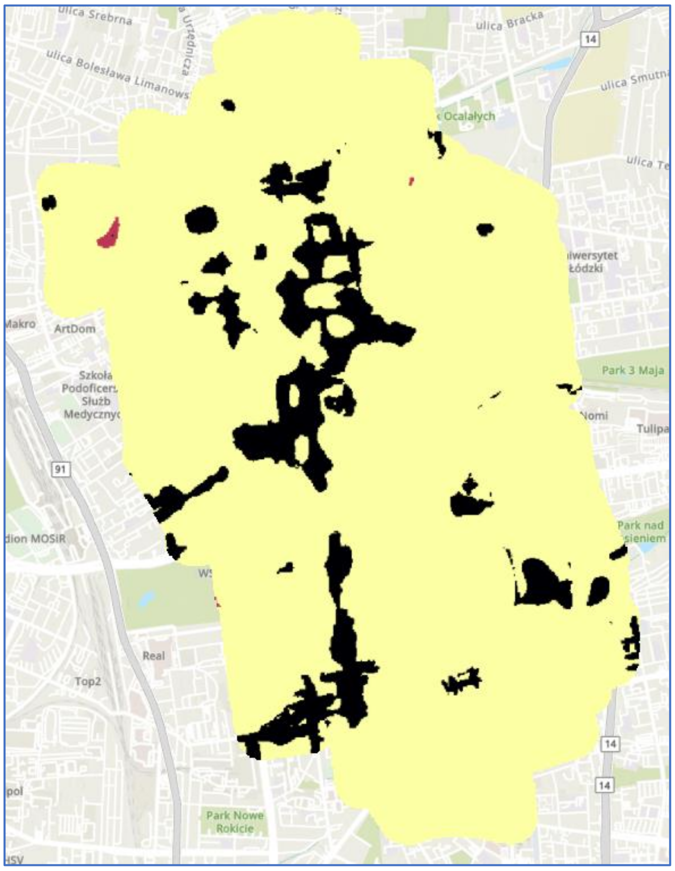


We connect all our criteria to the operation and after that;


Weighted Sum
Parameters
Environments
Properties
Input rasters
Raster K1
Field VALUE
Weight 0,2239
Raster K2_Distance_
Field VALUE
Weight 0,0833
Raster K3_Distance_
Field VALUE
Weight 0,1893
Raster K4_Intensity_
Field VALUE
Weight 0,3901
Raster K5_Residenti
Field VALUE
Weight 0,0524
Raster OWN_CRITEI
Field VALUE
Weight 0,0611
Add another
Output raster
Weighte_K11
OK


ulica Srebrna
ulica Bolesława Limanowskiego
ulica Bracka
ulica Smutna
Park 3 Maja
Tulipan
ArtDom
Szkoła Podoficerska Służb Medycznych
Park nad Jasieniem
Real
Top2
Park Nowe Rokicie
ulica Cieszyńska


Reclassify
Parameters
Environments
Properties
Input raster
Weighte_K1_D1
Reclass field
VALUE
Reclassification
Reverse New Values
Start
End
New
0,032425
0,07
1
0,07
0,4
2
NODATA
NODATA
NODATA
Classify
Unique
Output raster
Reclassify_Weights
Change missing values to NoData
OK

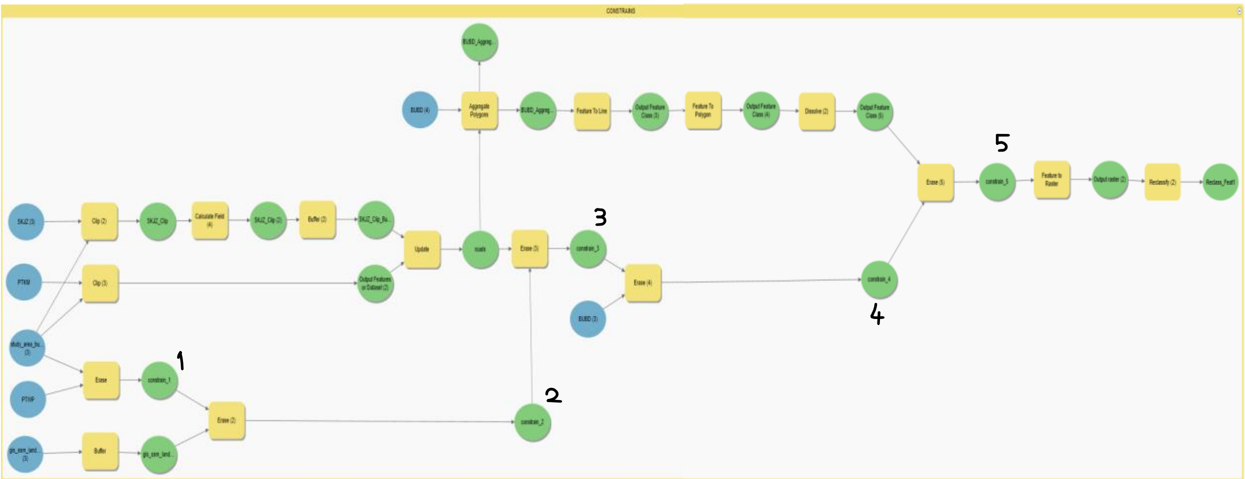

ulica Srebrna
ulica Bracka
ArtDom
Park 3 Maja
Real
Top2
Park Nowe Rokicie

We write our values to the matching criteria. And make a reclassify operation to see our final result. Once we get the final amount of information, we'll go on to the constraints procedure to figure out where we can't put these parcel lockers and find the most convenient locations. Constraints are components that enable the application of specific constraints during an analysis or optimization process. These limitations guarantee that the analysis's findings are applicable to actual situations and aid in producing outcomes that are practical. For instance, parcel locker positioning analysis enables consideration of constraints, crossroads, legal limitations, security limitations, or infrastructural needs. Results of analyses become more applicable and realistic as a result.

Our constraints are:

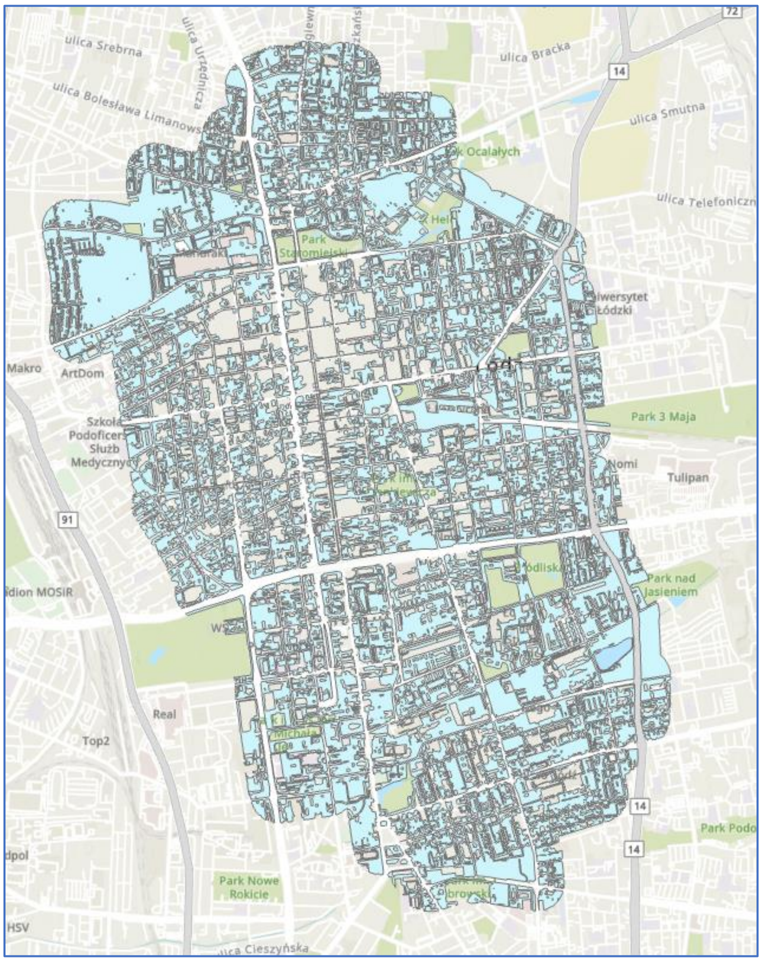
1) Not on the water bodies
2) Not in the middle of parks
3) Not on the roads
4) Not in the middle of the building
5) Not in the closed backyards

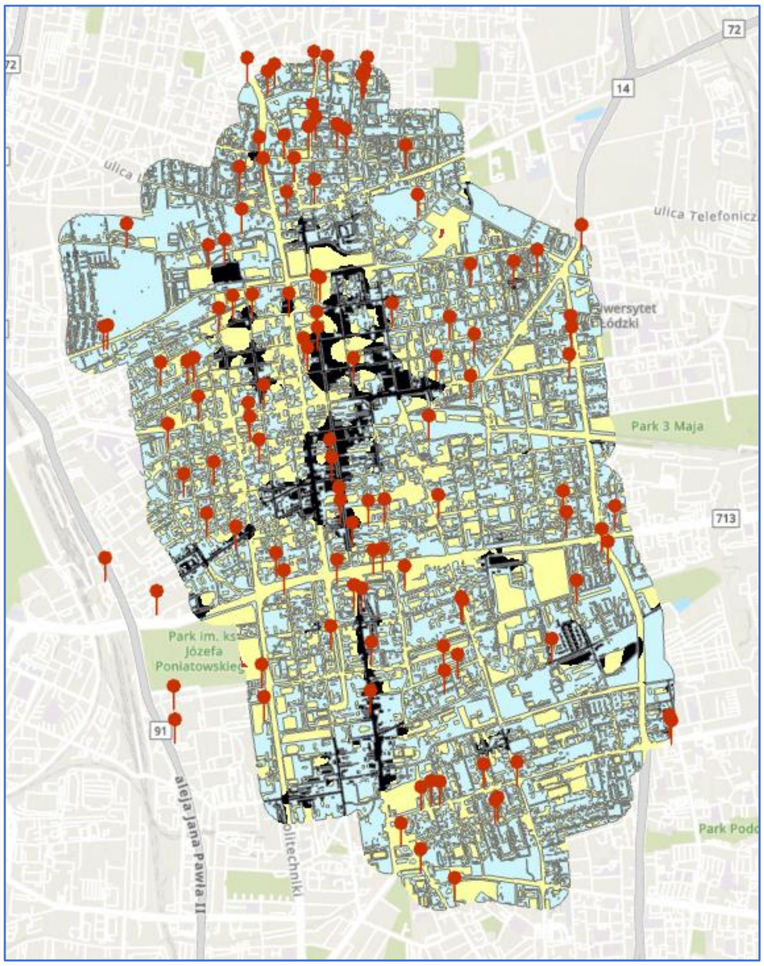
We built these constraints on the model builder by connecting them sequentially. Then we combined them all and got the result where we can't put them.



You can see the all operations in this model builder. Our result is:

After the constraints we can see the certain areas that shows us where we can put our parcel lockers. Now lets put together our reclassified weighted some result, parcel locker places and this constraints.





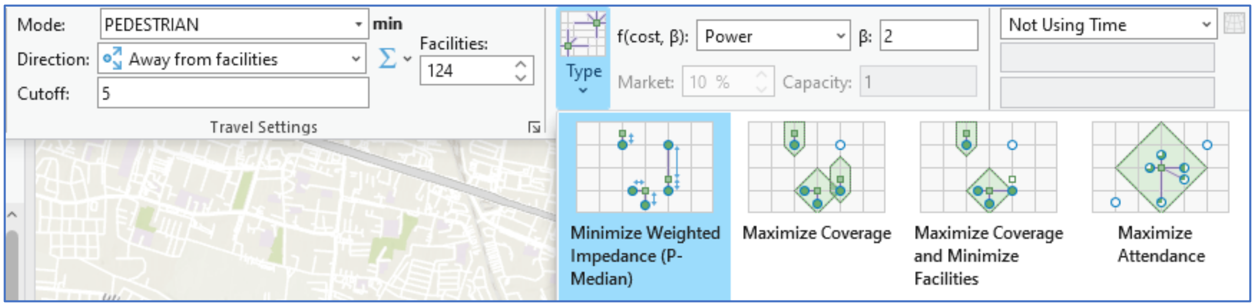
For example we can put our new parcel locker in a place away from other parcel lockers, taking into account the restrictions in this black area.

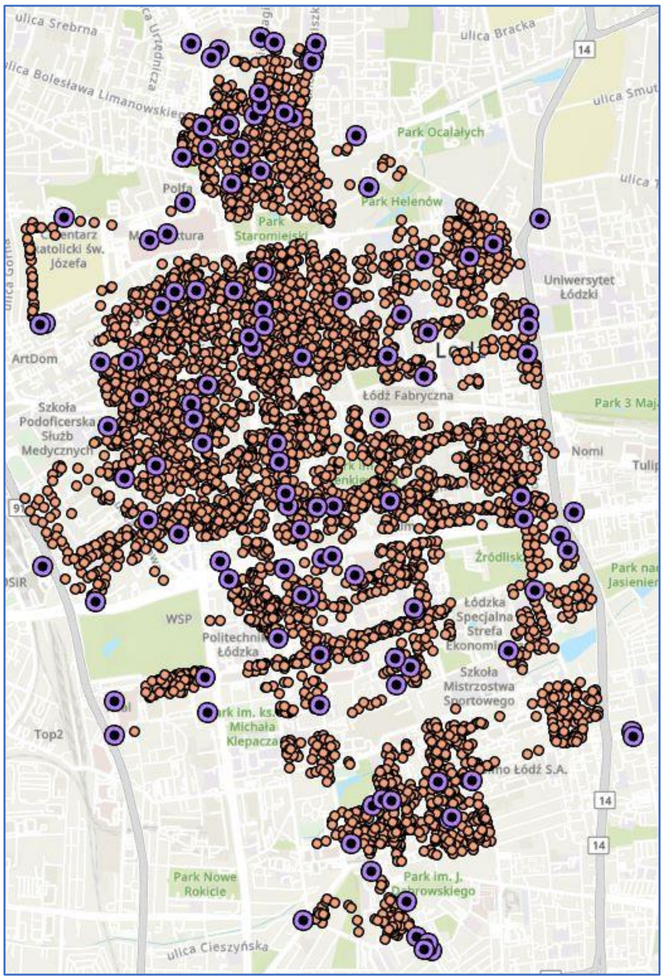
The network structures of geographic data can be understood and optimized via network analysis. It is applied to a variety of industries, including telecommunications, energy management, and planning for roads and transportation. This approach is helpful for figuring out the quickest or shortest routes, analyzing traffic flow, planning distribution, and figuring out communication coverage zones. With benefits like efficiency, cost savings, and sustainability, network analysis provides answers to issues that exist in the real world.

Placement of parcel lockers is intimately related to network analysis. Utilizing this analysis technique, parcel locker placement can be made as effective as possible. Utilizing geographic information, network analysis examines the current road system and identifies the best areas for user-friendly access. By maximizing connections between different parcel lockers, it also assists in determining the best distribution routes and the shortest distances. In order to promote the development of a quick and efficient distribution network, it is utilized as a key tool in network analysis, parcel locker layout and parcel locker network design.

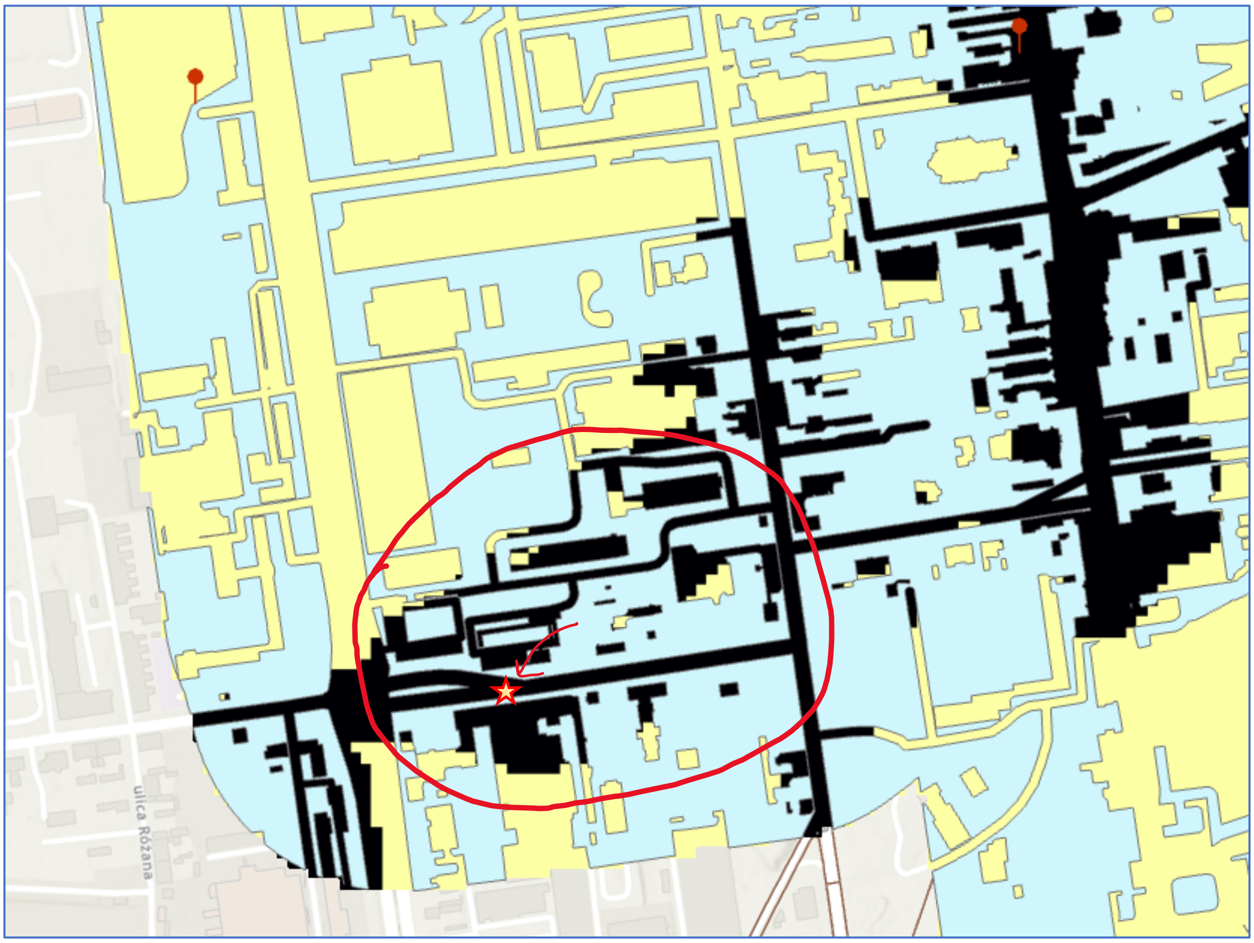
This is our network for this project;

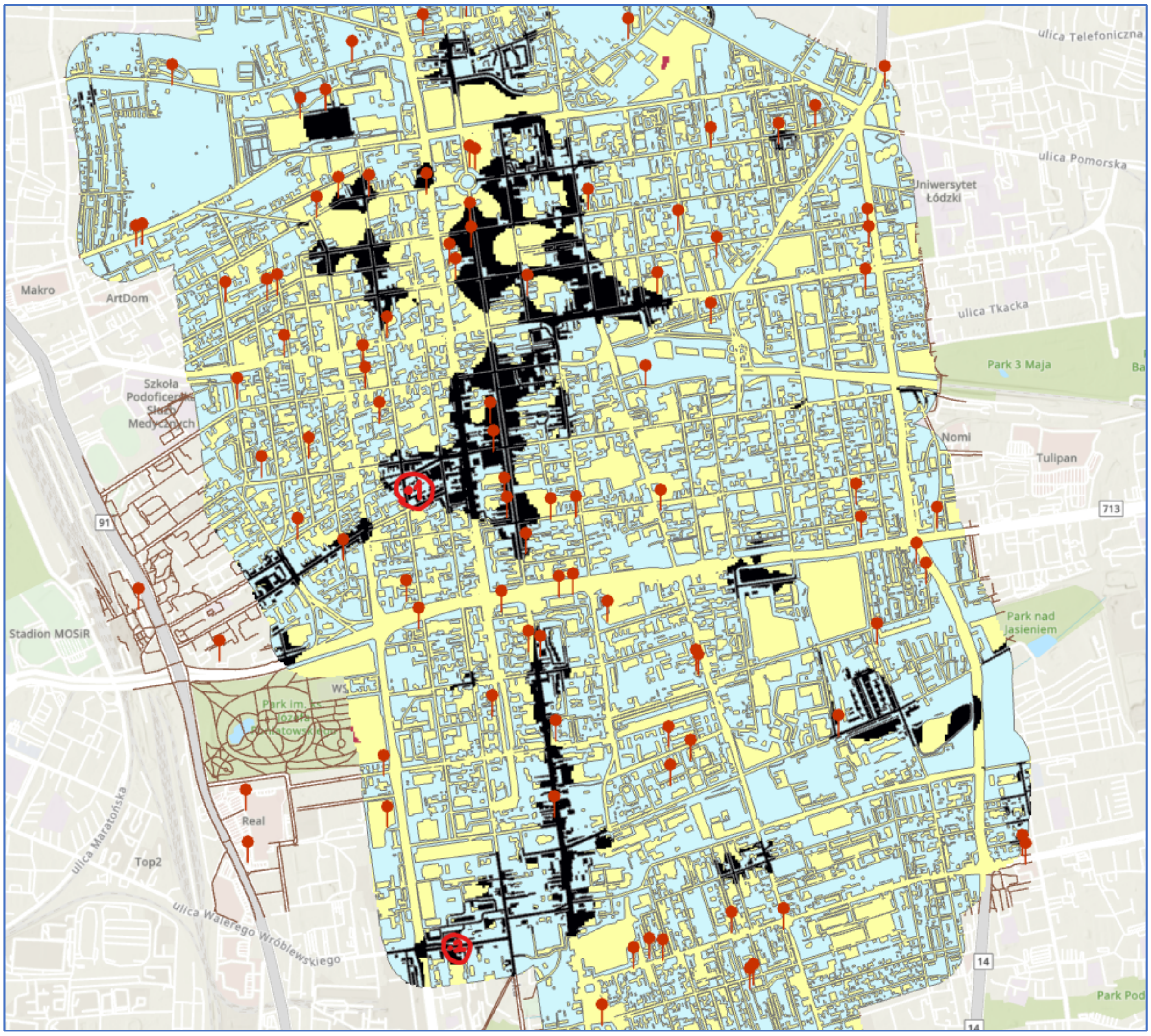
We use pedestrian paths for this network. After that we make a Location Allocation with inpost lockers, buildings etc. After that we perform a minimize weight operation.





After all this operations we can finally obtain some specific locations.


ulica Różana



These are our possible locations.

CONCLUSION

Several inferences can be made in light of the analysis done regarding parcel locker placements. The distance to roadways criterion, in particular, was found to be crucial for reducing delivery times and optimizing delivery routes. As a result, both couriers and recipients will have easy access. Second, choosing a site for parcel pickup that is convenient for customers and is close to park entrances might increase customer satisfaction. In order to reduce duplication and guarantee effective service area coverage, the distance to existing parcel lockers criterion also seeks to reduce overlap. Furthermore, taking Points of Interest (POI) into account can help make parcel lockers more visible and accessible, which will increase their use.

There are a number of factors to think about in order to better this kind of study. First, including more thorough and current data, such real-time traffic data, can lead to more accurate results. Second, adding more characteristics that are relevant to the neighborhood, including consumer preferences or population density, might enhance the research. Third, taking into account seasonal variations or dynamic demand-influencing factors, such holidays or special events, can result in a more resilient and adaptive solution. I want to open this special event or holiday event. When analyzing parcel locker locations, taking into account holidays or other notable events could provide valuable information that can help improve the delivery process. Due to internet shopping, giving, or event-related products, the amount of packages during holidays or special occasions usually increases. The algorithm can detect places with increased demand during particular times and allocate parcel lockers in accordance by taking these aspects into consideration. In addition to maintaining effective delivery operations during peak hours, this can help reduce congestion or overflow in specific places. Additionally, knowing how customers behave and what they prefer during holidays or other special occasions can be used to put parcel lockers in a way that best suits their demands and raises client happiness.

The analysis could benefit from including information on public transportation networks or pedestrian walkways as these can significantly affect accessibility. The criteria for a more customer-centric approach may also be improved with the help of feedback from stakeholders, such as couriers and recipients. The accuracy and efficiency of location planning for parcel lockers could be improved by a more thorough and dynamic analysis that incorporates several data sources and stakeholder input.